

1 Annotation Schema

We define two different main concepts of entities which are consumer food practice entities and climate change entities. Both of these concepts contain multiple subtypes of entities that have been manually annotated.

The first macro-type of entities are consumer food practices (actions directly carried out by an individual or a household as a food consumer). The subtypes of entities are defined according to the food itinerary of the consumer, from planning to waste management:

1. **Food repertoire:** Covers practices and factors related to food choices (in terms of quality or quantity) that may or may not be part of an individual's diet (e.g., meat consumption, food choices).
2. **Logistics and Travel:** Includes pre-purchase practices (checking existing resources, meal planning, etc.) and travel to the place of purchase (using low-carbon transportation to go grocery shopping)
3. **Purchasing:** Covers purchasing practices (product selection, choice of retail locations, interactions with marketing) as well as non-market procurement strategies (self-production).
4. **Storage:** Covers food storage practices after purchase and before preparation (choice of packaging, adherence to or disregard of expiration dates, preservation, etc.)
5. **Food preparation:** All attitudes and practices related to food preparation and cooking (e.g., raw food diets, water conservation, reducing energy consumption, minimizing waste during preparation, etc.). It also includes skills and actions that we perform when cooking.
6. **Consumption** (table manners): Covers consumption practices related to eating and meals (e.g., portion sizes, choice of utensils).
7. **Waste Management:** Covers practices that take place after a meal, with consideration for the end of a food item's life cycle (storing leftovers, composting, sorting food waste, expiration date reasoning...).

Climate change entities encompass the phenomena, factors, impacts, causes, and strategies associated with climate change, as described in the text or recognized as falling within the realm of climate. They include environmental events, greenhouse gases, climate impacts, anthropogenic causes of climate change, and actions aimed at mitigating or responding to its effects. Climate change entities may be identified using domain knowledge. However, any causal or relational link between these entities and food practices must be explicitly supported by the text. The subtypes of entities are defined according to entities selected by the expert annotator from the Climate-change-NER corpus [1], as well as an additional environmental concern type that was judged relevant by our expert:

- **Climate greenhouse gases:** Gases that cause heating of the atmosphere (e.g., "carbon dioxide", "CH₄", "GHG emission")
- **Climate hazards:** Dangers with potential negative impact on climate, such as floods, wildfires, droughts, and heatwaves.
- **Climate impacts:** Effects of hazards, primarily negative effects on humans. We also consider climate impacts on livestock as impacts, as it indirectly affects humans.
- **Climate problem origins:** Problems that describe why the climate is changing. Key examples are fossil fuel and deforestation. We also mention sectors that can be cited as causes of energy use. For instance, in a text about the energy consumption by the transport sector, the transport sector is annotated as a problem.

- **Climate mitigations:** Activities to reduce climate change or to better deal with the consequences.
- **Environmental concern:** This entity encompasses several aspects, e.g., pollution, biodiversity, waste, water, climate, etc.; it does not specifically refer to climate change; it is an intermediate/residual category within the environmental sphere for which the link to climate is not established or direct.

2 Annotation guidelines

In this section, we specify, in addition to section 1, the rules created by our annotator to help annotate the defined entity types in some specific and ambiguous cases. First, we define rules for specific types of entities, starting with food practices entities:

- For mentions of "behavior" related to food or diet, it should be annotated as "Food Repertoire" only if the text links these behaviors explicitly to climate change.
- Food practices must explicitly or implicitly involve a consumer. Macro-level phenomena, or those occurring downstream in the food chain (e.g., primary production, industrial processes), are not considered practices. The consumer must be involved directly when a food practice is annotated.
- Do not annotate mentions where we cannot identify a clear practice (e.g., consumer habits); too generic a term is not informative.
- Do not annotate food items (e.g., meat dishes, sustainable food) but take care of similar forms that refer to repertoire and that must be annotated (e.g., meat-free dishes).
- Food categories (e.g., wild foods, edible insects, organic foods) are not annotated as food practices unless they are embedded in a consumption or dietary context (e.g., consumption, diet, intake, eating patterns).
- Cognitive constructs are excluded unless they directly refer to operationalized food preparation activities or competencies immediately applied in food acquisition, preparation, or consumption.
- Include modifiers expressing intensity or direction when a practice is part of the observed behavioral change (e.g., increase/decrease).
- A practice that includes negation is still a practice and should be annotated (e.g., "Do not rinse meat", "Do not pay attention to expiration dates").

Then we also have rules for climate change entities :

- If a climate hazard is named in more detail in a text, the entire term is annotated (e.g., surface water flood or soil liquefaction).
- For the climate impact entity, vector-borne diseases and other diseases explicitly described as climate-sensitive are annotated as impacts. COVID-19 is annotated as an impact only when the text explicitly establishes a link between climate change and the pandemic.
- If an increase or decrease in activity reduces climate change, it should be annotated as a Climate Mitigation (e.g., "food waste has seen a notable decrease in volume").
- Transitions, strategies, policies, or transformations explicitly aimed at the sustainability of a system should be annotated as climate mitigation even when climate change is not explicitly mentioned.

- Environmental concern includes biodiversity, the ecological footprint, etc. Environmental concern includes generic or non-actionable references to sustainability, environmental awareness, or trends in consumption that do not describe explicit consumer actions.

2.1 Global rules

- Discontinuous annotation: If a practice is expressed as a predicate or attribute of diets (e.g., be varied, be seasonal), annotate the full conceptual unit of dietary recommendation using discontinuous spans including the implicit subject (e.g., diets should), and include all coordinated attributes as separate linked spans.
- Nested annotation: a food practice can have the same exact boundary as another entity type. For example, as some practices may have a mitigating effect on climate change, they can also be a climate mitigation entity. In the case of discontinuous mentions, we might even have some spans that have the same types for two different discontinuous mentions (but the other spans composing these mentions will be different).

References

- [1] Bishwaranjan Bhattacharjee, Aashka Trivedi, Masayasu Muraoka, Muthukumaran Ramasubramanian, Takuma Udagawa, Iksha Gurung, Nishan Pantha, Rong Zhang, Bharath Dandala, Rahul Ramachandran, Manil Maskey, Kaylin Bugbee, Mike Little, Elizabeth Fancher, Irina Gerasimov, Armin Mehrabian, Lauren Sanders, Sylvain Costes, Sergi Blanco-Cuaresma, Kelly Lockhart, Thomas Allen, Felix Grezes, Megan Ansdell, Alberto Accomazzi, Yousef El-Kurdi, Davis Wertheimer, Birgit Pfitzmann, Cesar Berrospi Ramis, Michele Dolfi, Rafael Teixeira de Lima, Panagiotis Vagenas, S. Karthik Mukkavilli, Peter Staar, Sanaz Vahidinia, Ryan McGranaghan, and Tsenggar Lee. INDUS: Effective and Efficient Language Models for Scientific Applications, October 2024.